Supplementary Material: A Bedside Decision Support System for Disorders of Consciousness: Multi-Scale Temporal Dynamics and Multicenter Validation

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SUPPLEMENTARY CLINICAL AND METHODOLOGICAL NOTES

A. CRRC Cohort Recruitment and Ethics Compliance

The prospective inpatient hospital cohort was evaluated at the China Rehabilitation Research Center (CRRC / Beijing Bo'ai Hospital, Capital Medical University Rehabilitation College). Inpatients presenting with disorders of consciousness secondary to severe acute brain injury (11 ICH, 4 TBI) were enrolled during neurorehabilitation. Ethical approval was granted by the Medical Ethics Committee of the China Rehabilitation Research Center (Approval No. 2025-118-01; cohort recruitment and auditory paradigm screening detailed in Duan et al. [1]). Written informed surrogate consent was obtained from legally authorized representatives prior to telemetry recording in accordance with the Declaration of Helsinki. Diagnostic categorization into minimally conscious state (MCS/EMCS, $N = 6$; 5 MCS and 1 EMCS) and vegetative state / unresponsive wakefulness syndrome (VS/UWS, $N = 9$) was established through serial Coma Recovery Scale-Revised (CRS-R) assessments by certified neurorehabilitation clinicians (scores: 2–23). Recordings were conducted during stable post-acute neurorehabilitation in patients free from continuous acute ICU sedation. Telemetry was acquired from the cohort in [1].

B. Telemetry Acquisition and Signal Preprocessing

Bedside electroencephalography was acquired using a research-grade NeuSen NSM2 amplifier / EEGCap32-pony system configured at a native sampling rate of 1,000 Hz with active Ag/AgCl electrodes positioned according to the international 10–20 montage (19 channels; see Duan et al. [1]). Continuous recordings spanned intermediate monitoring intervals (mean 19.53 ± 7.96 minutes; range 7.27 to 39.07 minutes). Raw continuous signals were re-referenced to Common Average Reference (CAR) and segmented into non-overlapping 4-second blocks (4,000 samples per block).

To prevent data leakage, artifact screening operated strictly at the individual recording level: blocks with peak-to-peak amplitude at or above $500\,\mu$V were flagged as artifactual gaps ($\mathbb{I}_{\text{adj}} = 0$). Homologous electrodes matching the discovery montage (F3, F4, C3, C4, O1, O2) were extracted. Two subjects (CRRC_008 and CRRC_009) exhibited persistent high-impedance drift isolated to the occipital channel O_2. To assess clinical robustness across sensor conditions, two complementary validation protocols were evaluated: (1) **Intact 6-Channel Subset** ($N = 13$): Excluded the two subjects with localized O_2 mechanical drift, evaluating records with fully intact 6-channel arrays; (2) **Full Prospective Cohort** ($N = 15$): Evaluated all 15 enrolled subjects to verify engineering tolerance against real-world single-channel sensor artifacts.

C. Matched Cross-Center Preprocessing and Operating Points

All matched comparisons use the same feature matrix and source training records for every model: $[c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]$. Source-only inner three-fold cross-validation selects ridge and logistic regularization; LDA uses analytical shrinkage. Full-source feature signs are $\mathbf{s}_{\text{source}} = [+1, +1, +1, -1]^\top$. Model fitting uses source-standardized, sign-coded coordinates.

Frozen-source transfer reuses source means, scales, and signs. Target-standardized transfer computes means and scales from the unlabeled target batch while retaining source signs. CRRC target-aligned transfer uses the same target-standardized matrix and $\mathbf{s}_{\text{target}} = [-1, +1, +1, +1]^\top$ for all models. In raw standardized coordinates, this corresponds to $\text{diag}(-1, +1, +1, -1)$ applied consistently to the target inputs of source-trained supervised models. Target labels enter only the final evaluation. The direction rule is specified for the subacute clinical setting; its prospective deployment requires the rule and calibration batch to be fixed before evaluating later patients. The primary four-feature score is the arithmetic mean; the fixed-truncation variant averages $\text{clip}(s_k z_k, -1.5, 1.5)$ under identical calibration (0.9291 AUROC discovery, 0.7963 full CRRC vs. 0.9324 and 0.7778 unclipped; Table S2). Primary decisions apply source-training score medians (CRRC truncated: sensitivity 0.6667, specificity 0.7778, balanced accuracy 0.7222; Figshare source-frozen threshold: balanced accuracy 0.5645, sensitivity 1.0000, specificity 0.1290). All transforms, traces, and input hashes are archived in results/matched_model_comparison_v1/; paired uncertainty applies 10,000 subject-level Dirichlet draws.

D. Section S-CDSS: Bedside Streamlined Implementation & Latency Profiling

To explore implementation feasibility, the four-feature temporal dynamic consensus pipeline was implemented as a streaming prototype (docs/bedside_cdss_demo.html, animation in docs/bedside_playback_demo.gif). The architecture accepts consecutive non-overlapping 4-s telemetry epochs ($f_s = 256\,\text{Hz}$), subjecting each block to artifact gating ($< 500\,\mu\text{V}$). Temporal continuity tracking ensures that artifact gaps break physical adjacency ($\mathbb{I}_{\text{adj}} = 0$); feature estimation is gated until ≥ 10 physically adjacent clean pairs accumulate. Data with

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TABLE S1
DE-IDENTIFIED INDIVIDUAL COHORT CHARACTERISTICS, FEATURE PROFILES, AND CROSS-CENTER BENCHMARK PERFORMANCE FOR THE CHINA REHABILITATION RESEARCH CENTER (CRRC / BEIJING BO'AI HOSPITAL) INPATIENT COHORT ($N = 15$; 15 MALES; 6 MCS [INCLUDING 1 EMCS], 9 VS/UWS; 11 ICH, 4 TBI; MEAN AGE 51.7 ± 12.3 YEARS). CLEAN 6-CHANNEL RECORDS ($N = 13$; 6 MCS [INCLUDING 1 EMCS], 7 VS/UWS) EXCLUDE CRRC_008 AND CRRC_009 DUE TO CONTINUOUS MECHANICAL ELECTRODE DRIFT ON O_2.

Subject ID	Diagnosis	CRS-R	Lesion Laterality	Duration (s)	Retention	c_{global}	$-\text{CV}_\theta$	ρ_1	r_{residual}
CRRC_001	MCS	11	Bilateral intact	605.0	98.7%	0.1834	-1.3927	+0.1118	+0.5878
CRRC_002*	MCS^{+}* (EMCS)	23	Right hemisphere	1236.0	98.7%	0.5343	-0.3331	+0.3201	+0.2064
CRRC_003	MCS	10	Left hemisphere	1225.0	97.1%	0.8409	-0.3682	+0.1451	+0.3646
CRRC_004	MCS	7	Left hemisphere	1221.0	100.0%	0.5993	-0.3748	-0.0378	+0.1391
CRRC_005	MCS	17	Bilateral intact	845.0	100.0%	0.4906	-0.6205	+0.3850	+0.3328
CRRC_006	MCS	9	Right hemisphere	1216.0	97.4%	0.5173	-0.4568	+0.5073	+0.2188
CRRC_007	VS/UWS	7	Right hemisphere	1896.0	99.8%	0.8119	-0.4731	+0.0105	+0.3776
CRRC_008†	VS/UWS	2	Bilateral intact	1213.0	100.0%	0.7253	-0.6384	+0.0337	+0.3048
CRRC_009†	VS/UWS	3	Bilateral intact	1209.0	99.7%	0.6213	-0.6091	+0.0614	+0.1660
CRRC_010‡	VS/UWS	2	Bilateral intact	637.0	21.4%	0.7294	-0.4525	+0.2401	-0.0657
CRRC_011	VS/UWS	7	Bilateral intact	1237.0	99.0%	0.4154	-0.4882	+0.4003	+0.3513
CRRC_012	VS/UWS	7	Bilateral intact	436.0	99.1%	0.5607	-0.3266	+0.0680	+0.0339
CRRC_013	VS/UWS	7	Left hemisphere	2344.0	99.5%	0.5517	-0.4399	+0.1056	+0.2702
CRRC_014	VS/UWS	7	Right hemisphere	1215.0	86.8%	0.7556	-0.6032	+0.6941	+0.1504
CRRC_015	VS/UWS	4	Left hemisphere	1040.0	100.0%	0.6218	-0.7362	-0.0033	+0.3001

*CRRC_002 met Emergence from MCS (EMCS) criteria; retained in conscious class (MCS^{+}) for binary triage. †Mechanical O_2 electrode drift; excluded in Protocol 1 ($N = 13$), retained in Protocol 2 ($N = 15$). ‡21.4% retention due to motion; retained under intent-to-treat. Telemetry: NeuSen NSM2 / EEGCap32-pony (1000 Hz, 4-s epochs, CAR, F3, F4, C3, C4, O1, O2; IRB No. 2025-118-01 [1]). Model comparisons across protocols appear in Table S2; operating thresholds are detailed in Note C.

TABLE S2
MATCHED-INPUT MODEL COMPARISONS (AUROC). ALL METHODS SHARE FEATURE COLUMNS, SOURCE-ONLY PARAMETER SELECTION, SCALING, DIRECTION CODING, AND SOURCE-TRAINING MEDIAN THRESHOLDS WITHIN EACH PROTOCOL. TRUNCATION IS A FIXED SCORING-RULE VARIANT ($c = 1.5$). TARGET STANDARDIZATION AND DIRECTION ALIGNMENT ARE UNLABELED BATCH ADAPTATIONS.

Cohort and protocol	Linear	Clipped	Ridge	Logistic	LDA
Mendeley, 25 outer folds	0.9324	0.9291	0.8813	0.8964	0.8769
CRRC 15, source frozen	0.5741	0.5741	0.5556	0.5556	0.5556
CRRC 15, target standardized	0.5000	0.5000	0.5185	0.5185	0.5370
CRRC 15, target aligned	0.7778	0.7963	0.7407	0.7593	0.7593
CRRC 13, target aligned	0.7143	0.7143	0.7143	0.6905	0.6905
Figshare 50, D_1, source frozen	0.7691	0.7606	0.7759	0.7674	0.7742
Figshare 50, D_1, target standardized	0.7657	0.7606	0.7708	0.7589	0.7691

TABLE S3
HYPERPARAMETER STABILITY: SOFT TRUNCATION AND T-GECF GATING GRID

(A) Soft Truncation Threshold c

Threshold	AUROC	PR-AUC	Bal. Acc.
$c = 1.0\sigma$	0.8958	0.8543	0.8080
$c = 1.5\sigma$ (Default)	**0.9149**	**0.8782**	**0.8399**
$c = 2.0\sigma$	0.9056	0.8662	0.8046
$c = 2.5\sigma$	0.8996	0.8622	0.8129
$c = 3.0\sigma$	0.8996	0.8622	0.8129
$c = \infty$ (Raw unconstrained)	0.8996	0.8622	0.8129

(B) T-GECF Gating Grid ($\theta_0 \times \kappa$)

	Sigmoid Gate Slope κ			
θ_0 (Quantile)	0.5	1.0	2.0	5.0
1.282 (90%)	**0.9220**	0.9182	0.9187	0.9149
1.645 (95%)	0.9160	0.9182	**0.9116**	0.9089
1.960 (97.5%)	0.9160	0.9149	0.9056	0.9056
2.326 (99%)	0.9127	0.9116	0.9022	0.8996
Ablate Cont. ($c = \infty$)	0.8996	0.8996	0.8996	0.8996

TABLE S4
CACHED-POWER REPLAY TIMINGS AND VERIFICATION (5,400 EPOCHS PER RECORD; TWO RECORDS)

Processing Phase	Calls/record	Mean (ms)	Mean p95 (ms)	Max (ms)
Cached Power Ingest	5,373	0.0098	0.0111	0.7090
Feature Extraction	27	2.1323	3.1117	3.9057
Consensus Scoring	27	0.0110	0.0123	0.0164
Evaluated Step Total	27	2.1542	3.1330	3.9261

Offline Consistency: Max feature $|\Delta| \leq 2.77 \times 10^{-13}$; Score error: 0.000.

< 10 adjacent pairs yield `INSUFFICIENT_DATA`. Source-frozen standardized coordinates ($z_k = s_k(x_k - \mu_k)/\sigma_k$) use archived thresholds ($\theta_0 = 0.0451$ for the raw score and $\theta_{1.5} = 0.0725$ for the clipped score), providing auditable bedside decision support.

Cached-power replay of two source-training records (5,400 epochs per subject; P25 [VS] and P36 [MCS]) showed numerical agreement with offline batch processing ($|\Delta_{\text{feature}}| \leq 2.77 \times 10^{-13}$, scoring error 0.000; Table S4). Cached-power replay timings were: cached power ingestion mean 0.0098 ms (p95 0.011 ms); four-feature extraction mean 2.13 ms (p95 3.11 ms); consensus scoring mean 0.011 ms; and total evaluated update mean 2.15 ms (p95 3.13 ms), well below the 4,000 ms inter-epoch window. Feature calculation and consensus scoring operate over 1,800× faster than the 4-s telemetry ingestion window. Table values average the two subject-level summaries (27 evaluated updates each), including their individual p95 values. Complete trajectories and test records are archived in results/bedside_playback_demo/.

REFERENCES

[1] F. Duan *et al.*, "Auditory paradigm-dependent EEG feature screening and representation for distinguishing consciousness states in male patients with disorders of consciousness," *Frontiers in Neurology*, vol. 17, p. 1927313, 2026.